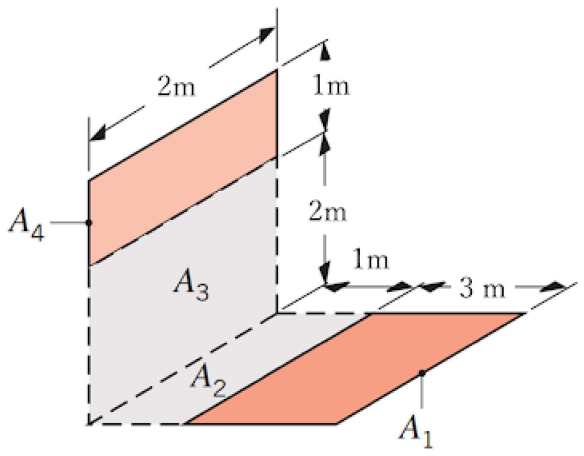
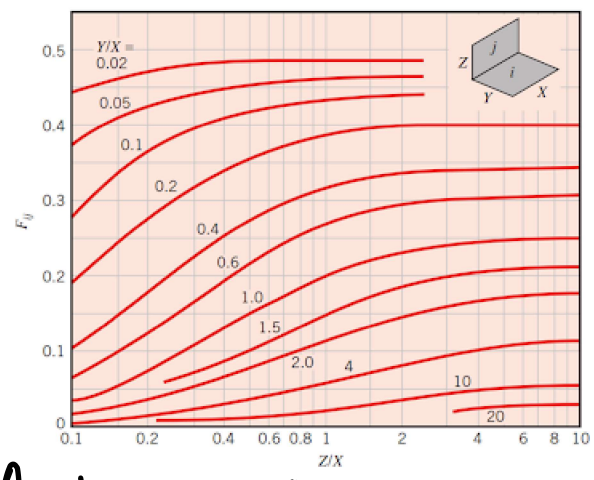
B-Energy Conversion MS4092

B-Energy Conversion MS4092

Number of problems: 8
Start at: 15.00 AM
Finish at: 15.23 AM
Duration: 23 Minutes

Surface 1 and 4 are perpendicular but do not share a common edge. Determine the view factor F14





$A_1 = 6m^2$
$A_2 = 2m^2$
$A_3 = 4m^2$
$A_4 = 2m^2$
$A_{12} = 8m^2$
$A_{34} = 6m^2$

$F_{12,34} = 0,14$
$F_{2,34} = 0,315$
$F_{12,3} = 0,12$
$F_{23} = 0,245$

- 0.05
- 0.03
- 0.04
- 0.02

$A_{12} F_{12,34} = A_1 F_{1,34} + A_2 F_{2,34}$
$F_{1,34} = 0,0817$
$A_{12} F_{12,3} = A_1 F_{13} + A_2 F_{23}$
$F_{13} = 0,0783$
$A_1 F_{1,34} = A_1 F_{12} + A_1 F_{14}$
$F_{14} = 0,0034$

In a convection phenomenon, the Nusselt number is proportional to

- Thermal conductivity
- Density
- Length
- Mass

$Nu = \frac{hL}{k}$

A Carnot cycle refrigerator is working between -15°C and 25°C. What is the COP?

- 6.45
- 7.45
- 7.2
- 25.8

$COP = \frac{Q_H}{Q_H - Q_C} = \frac{T_H}{T_H - T_C}$

$\downarrow$
W

$COP = \frac{298}{25-(-15)} = \frac{298}{40} = 7,45$

$T_H = 25 + 273$
$= 298\ k$

$$\eta = 1 - \frac{1}{r^{k-1}}\left[\frac{r_c^k - 1}{k(r_c - 1)}\right]$$

A Diesel engine operates with an ideal cycle that has a compression ratio of 17 and a cutoff ratio of 1.2. If it produces 140 kW, rate of heat addition to the cycle?

- 207.2 kW
- 211.5 kW
- 210.3 kW
- 212.2 kW

$r = 17 \quad k = 1,4$

$r_c = 1,2$

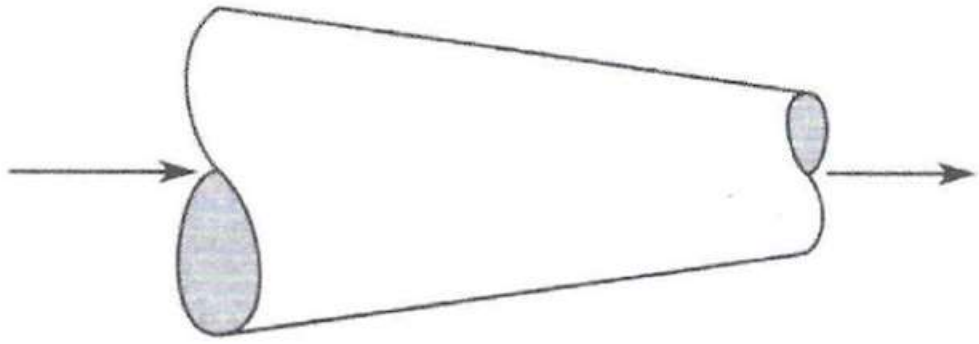
$\eta = 1 - \frac{1}{17^{1,2-1}}\left[\frac{1,2^{1,4}-1}{1,4(1,2-1)}\right] = 0,407$

$\eta = 41,07\%$

$\eta = \frac{W}{Q}$

$Q = 340,9\ kW$

A converging nozzle with inlet diameter of 150 mm and outlet diameter of 75 mm shown as follow, has the flowing water at inlet with a velocity of 1.1 m/s and discharged to the atmosphere. With the water is incompressible and friction effect is negligible, what is the gage pressure at the inlet?



- 11.3 kPa
- 10.8 kPa
- 12.7 kPa
- ~~9.1 kPa~~

$A_1 V_1 = A_2 . V_2 \qquad P_2 = 10^5\ Pa$

$V_2 = V_1 . \frac{D_1^2}{D_2^2}$

$V_2 = 1,1 . \frac{150^2}{75^2}$

$V_2 = 4,4\ m/s$

$P_1 + \cancel{\gamma z_1} + \frac{1}{2}\rho V_1 = P_2 + \cancel{\gamma z_2} + \frac{1}{2}\rho V_2^2$

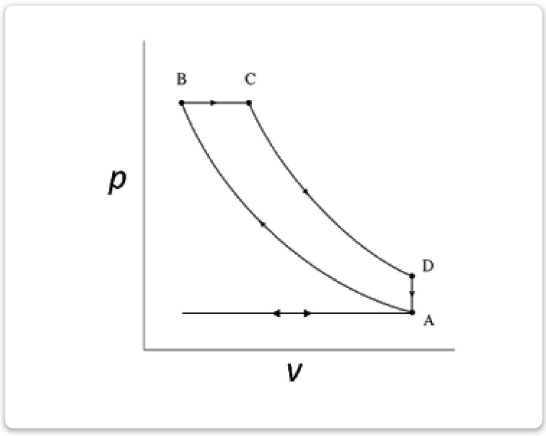
$P_1 = 10^5 + \frac{1}{2} . 1000\left(4,4^2 - 1,1^2\right)$

$P_1 = 109,075\ kPa$

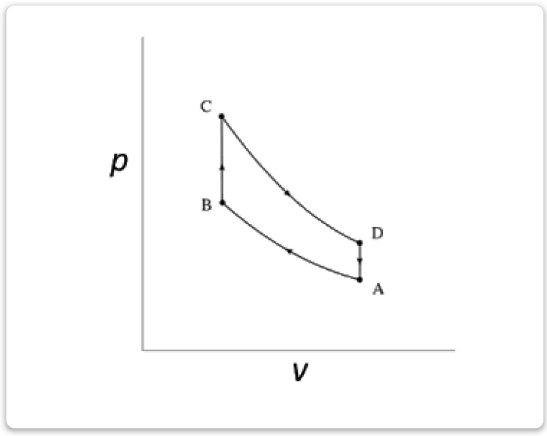
$P_{1g} = P_1 - 100\ kPa$

$P_{1g} = 9,075\ kPa$

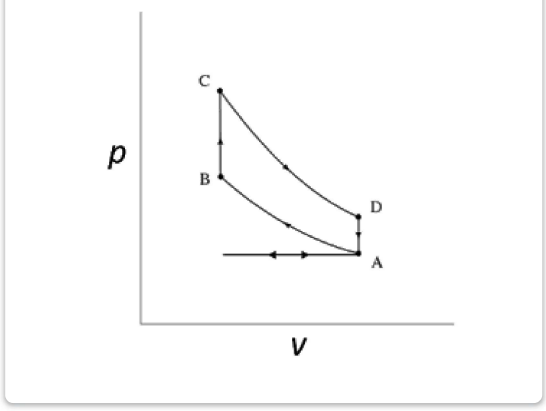
Which is the following diagram shows Diesel cycle?



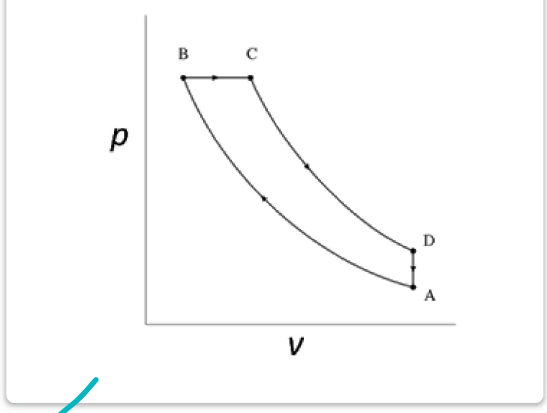
Option 2



Option 1



Option 4



Option 3

Yang lain:

A condenser of AC has surface temperature of 40°C. If the condenser surface is considered as black body, what is the heat flux emitted by the condenser surface? (Stefan-Boltzmann constant = 5.67×10^-8 W.m-2.K-4)

$T = 40°C + 273 = 313\ K$

$\varepsilon \sigma T^4 = 5,67 \times 10^{-8} \times 1 \times 313^4$

$= 544.2\ W/m^2$

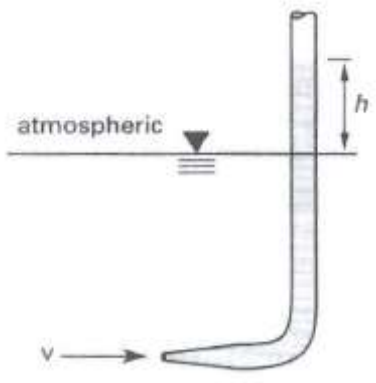
617.1 W/m2

698.5 W/m2

544.2 W/m2

786.2 W/m2

What is the height of water in the pitot tube, if the water's velocity flowing in the pipe is 3 m/s (g = 9.81 m/s2)



$\frac{1}{2}\rho V^2 = \rho g h$

$h = \frac{V^2}{2g}$

$= \frac{3^2}{2(9,81)}$

$h = 0.46\ m$

0.2

0.46

0.82

1.27

Kembali Kirim

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